

Relations

11/12/2013

A category $\mathcal{E}$ is regular when

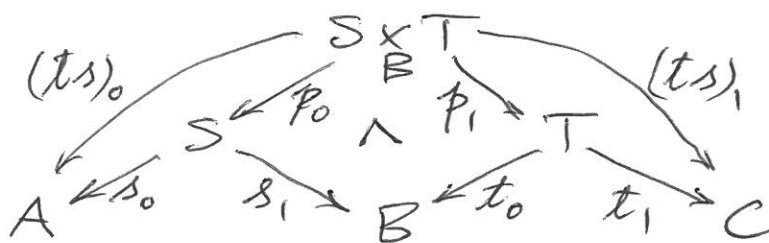
- all finite limits exist
- every morphism f factors as $f = me$ with m mono and e strong epi;
- every pullback of a strong epi is a strong epi.

In other words, (strong epi, mono) factorizations exist and are stable under pullback.

Let $\text{Spn } \mathcal{E}$ be the usual bicategory of spans in $\mathcal{E}$; the objects are those of $\mathcal{E}$ and morphisms $s: A \rightarrow B$ are spans

$$A \xleftarrow{s_0} S \xrightarrow{s_1} B$$

in $\mathcal{E}$. Horizontal composition is by pullback.



There is a normal lax functor

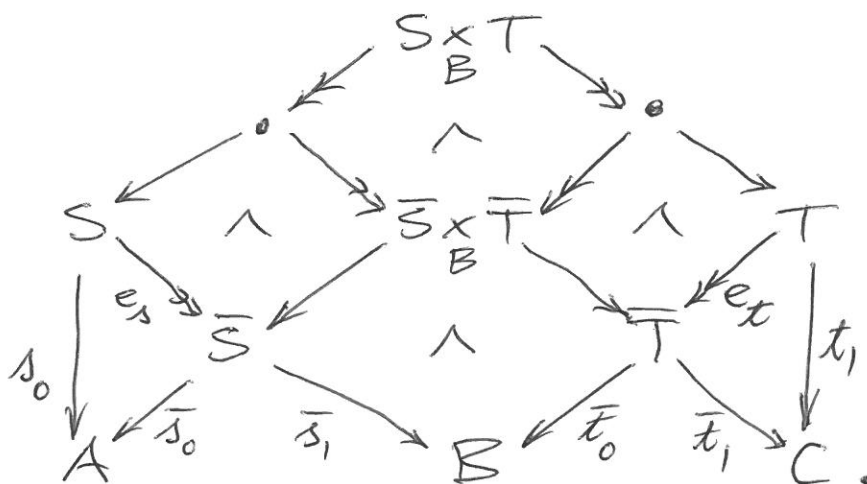
$$[-] : \text{Spn } \mathcal{E} \longrightarrow \text{Spn } \mathcal{E}$$

Rel ②

which is locally an idempotent monad; it is the identity on objects. For a span $s: A \rightarrow B$, define $\bar{s}: A \rightarrow B$ by

$$\begin{array}{ccc} s & \xrightarrow{(s_0, s_1)} & A \times B \\ & \searrow e_s & \nearrow (\bar{s}_0, \bar{s}_1) \\ & \bar{s} & \end{array}$$

where we write $\twoheadrightarrow$ for strong epis and $\rightarrow$ for monos. For composable spans $s: A \rightarrow B$, $t: B \rightarrow C$, form the diagram:



We obtain $\varphi: \bar{t} \circ \bar{s} \rightarrow \overline{t \circ s}$ as in the diagram:

$$\begin{array}{ccccc} s \times_B t & \xrightarrow{\quad} & \bar{s} \times_B \bar{t} & \xrightarrow{\quad} & \overline{s \times_B t} \\ \downarrow & \searrow \varphi & \downarrow & & \downarrow \\ \overline{s \times_B t} & \xrightarrow{\quad} & A \times C & \xleftarrow{\quad} & \end{array}$$

This gives the normal lax functor structure.

Rel(3)

By uniqueness of factorization, notice from the right-hand triangle in the last diagram, we have an isomorphism

$$\overline{S \times_B T} \cong \overline{S \times T};$$

this yields the invertibility of

$$\overline{\varphi} : \overline{t \circ s} \longrightarrow \overline{t \circ s}.$$

This allows us to apply Brian Day's reflection theorem for bicategories.

$$\text{Rel } \mathcal{E} \xrightleftharpoons[\perp]{(\overline{\quad})} \text{Spn } \mathcal{E} \quad (*)$$

We obtain a bicategory $\text{Rel } \mathcal{E}$ with the same objects as $\mathcal{E}$, with morphisms

$r : A \longrightarrow B$ those $\text{span}_{\mathcal{E}} e : r \implies \overline{r}$ ^{for which} $\overline{r}$ is invertible,

and composition defined by

$$r' \circ_{\text{Rel}} r = \overline{r' \circ_{\text{Spn}} r}.$$

The main point is associativity of composition (I'll write $\circ$ for $\circ_{\text{Spn}}$):

$$(r'' \circ_{\text{Rel}} r') \circ_{\text{Rel}} r = \overline{(r'' \circ_{\text{Rel}} r') \circ r} \cong \overline{r'' \circ r' \circ r} \cong \overline{r'' \circ r' \circ r},$$

where $r'' \circ r' \circ r$ denotes the span triple component, ^{Rel (4)} and the result follows by symmetry.

In fact, $(*)$ is a local (or lax) adjunction: $\text{Rel } \mathcal{E} \longrightarrow \text{Spn } \mathcal{E}$ is the identity on objects and a local inclusion; it is a normal lax functor while its local adjoint $(\bar{})$ is a pseudofunctor.

The spans $r: A \rightarrow B$ which are in $\text{Rel } \mathcal{E}$ are those which are jointly mono:

$$R \xrightarrow{(r_0, r_1)} A \times B.$$

In particular, for each $f: A \rightarrow B$ in $\mathcal{E}$, the spans $f_*: A \rightarrow B$ and $f^*: B \rightarrow A$ respectively defined as

$$A \xleftarrow{1} A \xrightarrow{f} B \quad \text{and} \quad B \xleftarrow{f} A \xrightarrow{1_A} A$$

are clearly jointly mono. So we have pseudofunctors

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{()_*} & \text{Spn } \mathcal{E} \\ ()_* \searrow & & \nearrow \\ & \text{Rel } \mathcal{E} & \end{array} \qquad \begin{array}{ccc} \mathcal{E}^{\text{op}} & \xrightarrow{()^*} & \text{Spn } \mathcal{E} \\ ()^* \searrow & & \nearrow \\ & \text{Rel } \mathcal{E} & \end{array},$$

all the identity on objects. The pseudofunctor

Rel ⑤

$(\bar{}) : \text{Spn } \mathcal{E} \rightarrow \text{Rel } \mathcal{E}$ takes the adjunction $f_* \dashv f^*$ in $\text{Spn } \mathcal{E}$ to an adjunction $f_* \dashv f^*$ in $\text{Rel } \mathcal{E}$; since $\text{Rel } \mathcal{E}$ is locally ordered, this last amounts to

$$f_* f^* \leq 1 \quad \text{and} \quad 1 \leq f^* f_*$$

It can be shown that $r : A \rightarrow B$ has a right adjoint in $\text{Rel } \mathcal{E}$ if and only if $r_0 : R \rightarrow A$ is invertible; this implies

$r \cong (r, r_0^{-1})_*$. We can thus recapture the category $\mathcal{E}$ from the bicategory $\text{Rel } \mathcal{E}$ by taking the isomorphism classes of left adjoint morphisms as morphisms.

Proposition (a) $f : A \rightarrow B$ in $\mathcal{E}$ is mono iff $f_* : A \rightarrow B$ in $\text{Rel } \mathcal{E}$ is a coretraction.

(b) $f : A \rightarrow B$ in $\mathcal{E}$ is strong epi iff $f_* : A \rightarrow B$ in $\text{Rel } \mathcal{E}$ is a retraction.

Proof (a) $r f_* = 1$ in $\text{Rel } \mathcal{E}$ implies $f^* = r f_* f^* \leq r$, so $1 \leq f^* f_* \leq r f_* = 1$. Clearly $f^* f_* = 1$ implies f mono, and conversely.

(b) Similarly $f_* r = 1$ implies $1 = f_* f^*$ which

is equivalent to $\mathcal{S}$ strong epi. $\square$

First Theorem of Category Theory In a regular category, every strong epi is a coequalizer.

Proof Let $e: X \twoheadrightarrow Y$ in $\mathcal{E}$ and form the kernel pair $u, v: K \rightarrow X$. We need to show that e is the coequalizer of u, v .

$$\begin{array}{ccccc} K & \xrightarrow{u} & X & \xrightarrow{e} & Y \\ & \xrightarrow{v} & & & \\ & & \searrow f & & \\ & & Z & & \end{array}$$

Suppose $fu = fv$. We have a span $Y \xleftarrow{e} X \xrightarrow{f} Z$ from Y to Z , but we want a morphism $w: Y \rightarrow Z$ in $\mathcal{E}$. We can use $(\bar{})$ to obtain a relation $r: Y \rightarrow Z$; that is, we have

$$\begin{array}{ccc} X & \xrightarrow{(e, f)} & Y \times Z \\ \searrow s & & \nearrow (r_0, r_1) \\ & R & \end{array}$$

We need to prove r_0 is invertible. Since $r_0 s = e$ is strong epi, so too is r_0 . It remains to prove r_0 is mono. Let $m, n: H \rightarrow R$ be the kernel pair of r_0 .

I claim we have a pullback

$$\begin{array}{ccc} K & \xrightarrow{t} & H \\ (u,v) \downarrow & & \downarrow (m,n) \\ X \times X & \xrightarrow{s \times s} & R \times R \end{array} \quad (*)$$

Since $r_0 s u = e u = e v = r_0 s v$, we obtain t from the kernel pair property of (m, n) . To prove the pullback property, take

$$\begin{array}{ccc} W & \xrightarrow{a} & H \\ (b,c) \downarrow & & \downarrow (m,n) \\ X \times X & \xrightarrow{s \times s} & R \times R \end{array}$$

Then $e b = r_0 s b = r_0 m a = r_0 n a = r_0 s c = e c$ implies there exists a unique $d: W \rightarrow K$ such that $b = u d$ and $c = v d$. Also $(m, n) t d = (s u, s v) d = (s \times s)(b, c) = (m, n) a$; so $t d = a$. So $(*)$ is a pullback. That t is epi will follow from the Lemma below. So $m t = s u = s v = n t$ implies $m = n$. So r_0 is mono. So r_0 is invertible. Put $w = r_1 r_0^{-1}$ and we have $w e = r_1 r_0^{-1} e = r_1 r_0^{-1} r_0 s = r_1 s = f$ as required, $\square$

Lemma In a regular category, if $e_1: X_1 \rightarrow Y_1$ and $e_2: X_2 \rightarrow Y_2$ are strong epi then so is $e_1 \times e_2: X_1 \times X_2 \rightarrow Y_1 \times Y_2$.

Proof A composite of strong epis is strong epi so it suffices to prove $e_1 \times 1$ and $1 \times e_2$ are strong epis. However, we always have the pullbacks

$$\begin{array}{ccc} X \times Z & \xrightarrow{e \times 1} & Y \times Z \\ \downarrow \pi_1 & & \downarrow \pi_1 \\ X & \xrightarrow{e} & Y \end{array};$$

so $e \times 1$ is a pullback of a strong epi. $\square$

A relation $r: A \rightarrow A$ on A is an equivalence relation when $1_A \leq r$, $r \circ r \leq r$ and $r^\circ \leq r$ in $\text{Rel } \mathcal{E}$. (Here $r^\circ = (B \xleftarrow{r_1} R \xrightarrow{r_0} A)$ for any $r: A \rightarrow B$ in $\text{Rel } \mathcal{E}$.)

A category $\mathcal{E}$ is Barr-exact when, for every equivalence relation $r: A \rightarrow A$, there exists a morphism $f: A \rightarrow B$ such that (r_0, r_1) is a kernel pair for f .

Rel(9)

It follows that, for every equivalence relation $r: A \rightarrow A$, the pair (r_0, r_1) has a coequalizer; it is obtained as the strong epi in factorizing an f as in the definition. Myles Tierney proved

$$\boxed{\text{Barr-exact} + \oplus \text{additive} = \text{abelian}}.$$

However this "+" is not cancellative: we also have

$$\boxed{\text{Puppe-exact} + \oplus \text{additive} = \text{abelian}}.$$

Puppe-exact categories were called "exact categories" in the Barry Mitchell book.

They are pointed categories and are generally not finitely complete; the concept is self dual.

A category $\mathcal{C}$ is pointed when it is enriched in the category $\mathbf{1}/\mathbf{Set}$ of pointed sets. There are morphisms $0: A \rightarrow B$ for every A, B , and $g \circ f = 0$.

Rel (10)

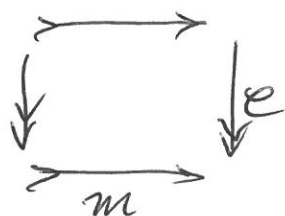
The kernel of a morphism is the equalizer of it with 0. The cokernel of a morphism is the coequalizer with 0. A morphism is a normal mono when it is a kernel of some morphism. A morphism is a normal epi when it is a cokernel.

A Puppe-exact category is a pointed category in which every morphism factors as a normal epi followed by a normal mono. It follows that every morphism has a kernel and a cokernel. Also every mono is normal and every epi is normal.

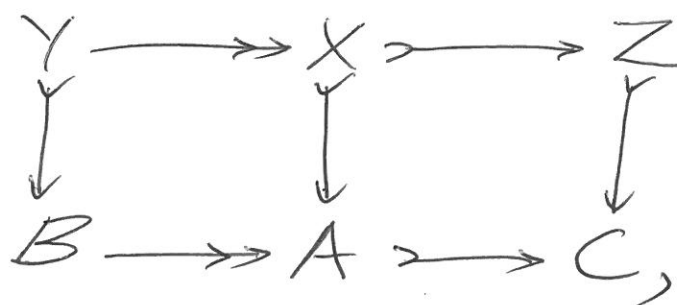
Pullbacks of monos with any morphism exist: to form the pullback of $U \xrightarrow{m} Y$ and $f: X \rightarrow Y$, we take any $u: Y \rightarrow Z$ for which m is the kernel of u ; the desired pullback is $\ker(uf)$:

$$\begin{array}{ccccc} \ker(uf) & \xrightarrow{\quad} & X & & \\ \downarrow & \wedge & \downarrow f & & \\ U & \xrightarrow{m} & Y & \xrightarrow{u} & Z. \end{array}$$

Actually, we can construct pushouts of epis with any morphism. A square



is a pullback if and only if it is a pushout. A pullback of an epi along a mono is an epi. A pushout of a mono along an epi is a mono. Moreover, in the diagram



the outside rectangle is a pullback if and only if the two squares are pullbacks.

Examples of Opre-exact categories

- 1) Every abelian category: modules over a ring, module-valued sheaves on a site.
- 2) Every non-empty full subcat. of a Opre-exact category which is closed under subobjects and quotients: cyclic groups,

vector spaces of $\dim \leq n$.

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3) The category Mlc of modular lattices and modular connections. A lattice A is modular when $x \leq b$ implies

$$x \vee (a \wedge b) = (x \vee a) \wedge b.$$

A connection is an order adjunction

$$A \begin{array}{c} \xrightarrow{f_*} \\ \perp \\ \xleftarrow{f^*} \end{array} B$$

with $f^* f_*(a) = a \vee f^*(0)$ and $f_* f^*(b) = b \wedge f_*(1)$.

Every Puppe-exact category $\mathcal{E}$ has the functor

$$\text{Sub} : \mathcal{E} \longrightarrow \text{Mlc}.$$

The full subcategory Dlc ^{of Mlc consisting} of distributive lattices is also Puppe-exact; in this case f_* preserves binary meets and f^* preserves binary joins.

4) The category $\mathcal{I}$ of sets and injective partial functions.

$$X \setminus U \xrightarrow{i_*} X \xrightarrow{(U, f)} Y \xrightarrow{j^*} Y \setminus \text{im } f$$

$$(U, f) \searrow \text{im } f \nearrow$$

$$X \setminus U \hookrightarrow X$$

$$Y / \text{im } f \xrightarrow{j} Y$$

Actually,

Puppe-exact + binary products = abelian

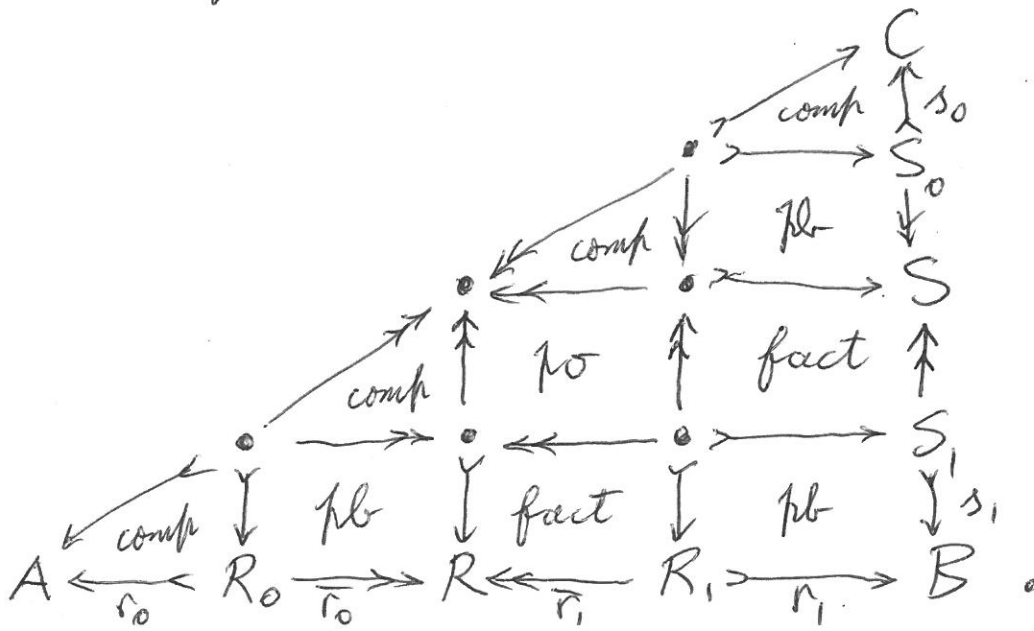
The bicategory $\text{Rel } \mathcal{E}$ of relations construction for a Puppe-exact category $\mathcal{E}$ is quite different from the regular case. (Brinkmann and Puppe [SLNM 96]¹⁹⁶⁹ point out that it also works for the category Grp of groups.)

As before the objects of $\text{Rel } \mathcal{E}$ are those of $\mathcal{E}$.

A morphism $r: A \rightarrow B$ is a diagram

$$A \xleftarrow{r_0} R_0 \xrightarrow{\bar{r}_0} R \xleftarrow{\bar{r}_1} R_1 \xrightarrow{r_1} B$$

in $\mathcal{E}$. The 2-cells are the diagram morphisms (fixed at A and B). The composition goes back to [M. S. Tsahlenko, Dokl. Acad. Nauk. SSSR 155 (1964); Math. Sbornik. 73 (1967)] and is as follows: for $A \xrightarrow{r} B \xrightarrow{s} C$ in $\text{Rel } \mathcal{E}$, form



Amazingly, this is associative! I would like a nice proof of this as in the regular case.

There is an obvious reversal $()^0 : (\text{Rel } \mathcal{E})^{\text{op}} \rightarrow \text{Rel } \mathcal{E}$ defined by

$$(A \leftarrow R_0 \rightarrow R \leftarrow R_1 \rightarrow B)^0 = (B \leftarrow R_1 \rightarrow R \leftarrow R_0 \rightarrow A).$$

We have a pseudofunctor

$$()_* : \mathcal{E} \longrightarrow \text{Rel } \mathcal{E}$$

which is the identity on objects and takes $f : A \rightarrow B$ to

$$A \xleftarrow{1} A \xrightarrow{e} \bullet \xleftarrow{1} \bullet \xrightarrow{m} B$$

where $f = m \circ e$. Also $f^* = (f_*)^0$.

For $r : A \rightarrow B$ in $\text{Rel } \mathcal{E}$, we have

$$r \circ^0 r \cong r.$$

By duality, $r^0 \circ r^0 \cong r$, so r^0 is a "generalized inverse" for r .

A reference for this is above material is

[Marco Grandis, "Homological Algebra: The interplay of homology with distributive lattices and orthodox semigroups (World Scientific, 2012)].

There is a functor

$$\bar{(\cdot)} : \text{Spn } \mathcal{E}(A, B) \longrightarrow \text{Rel } \mathcal{E}(A, B)$$

taking $s : A \rightarrow B$ to $A \xleftarrow{\quad} \bar{S} \xrightarrow{\quad} B$ where



The problem is that $\mathcal{E}$ has insufficient pullbacks to assure that $\text{Spn } \mathcal{E}$ is a bicategory in the usual way. Let $\tilde{\mathcal{E}}$ denote the full subcategory of $[\mathcal{E}^{\text{op}}, 1/\text{Set}]$ consisting of the pointed functors which preserve monos and preserve pullbacks of monos along arbitrary morphisms. The restricted Yoneda embedding

$$\gamma : \mathcal{E} \longrightarrow \tilde{\mathcal{E}}, \quad A \longmapsto \mathcal{E}(-, A)$$

is fully faithful, preserves any limits that exist in $\mathcal{E}$, and preserves epis and their pushouts. Then we have a functor

$$i : \text{Rel } \mathcal{E}(A, B) \longrightarrow \text{Spn } \tilde{\mathcal{E}}(A, B)$$

$$\text{taking } r : A \xleftarrow{\bar{r}_0} R_0 \xrightarrow{\bar{r}_0} R \xleftarrow{\bar{r}_1} R_1 \xrightarrow{\bar{r}_1} B$$

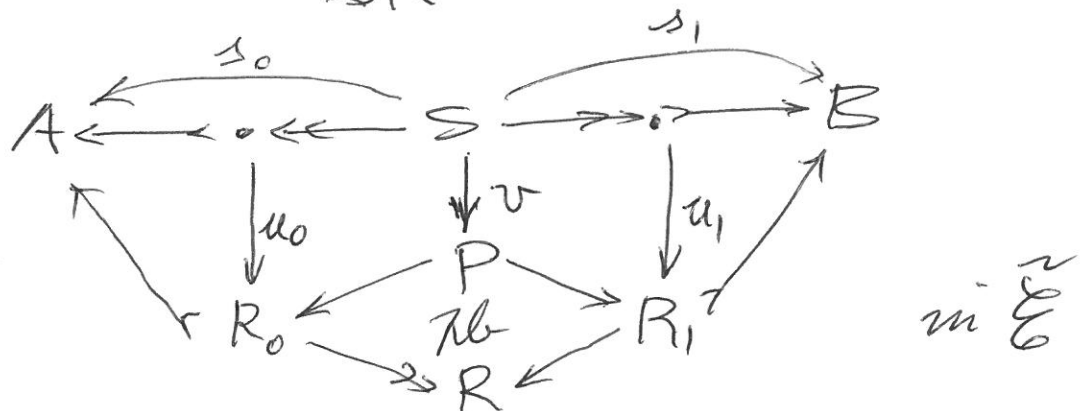
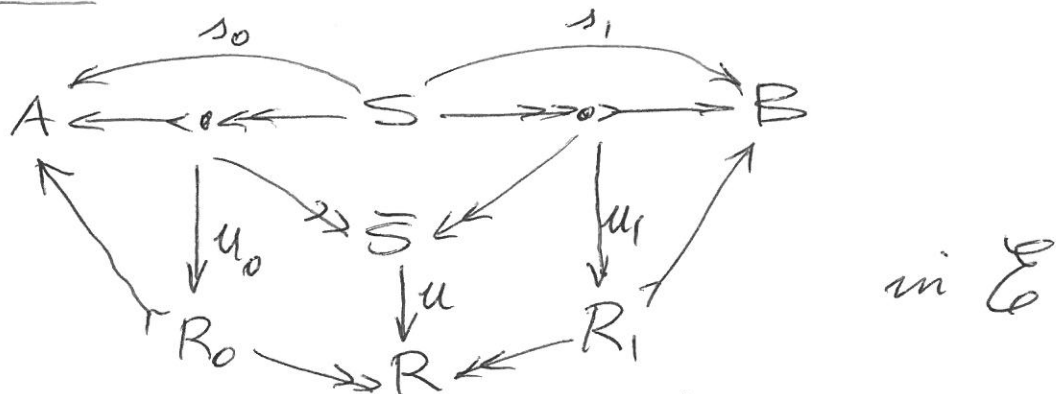
Rel (16)

to the span $A \xleftarrow{r_0 p_0} P \xrightarrow{r_1 p_1} B$ where

$$\begin{array}{ccccc} P & \xrightarrow{p_1} & R_1 \\ p_0 \downarrow & p_0 & \downarrow \bar{r}_1 \\ R_0 & \xrightarrow{\bar{r}_0} & R \end{array} \text{ in } \mathcal{E}.$$

Proposition $\text{Rel } \mathcal{E}(A, B)(\bar{s}, r) \cong \text{Spn } \tilde{\mathcal{E}}(A, B)(1, i(r))$

Proof



Notice that u_0 and u_1 are uniquely determined given

